SIMATS ENGINEERING

ASSESSMENT TOOL 2: Schema Analysis and Viva Voce

COURSE NAME: Query Processing for Data Science **COURSE CODE:** DSA0503

COURSE OUTCOME COVERED

CO2: Use Python basics and SQL libraries to connect to databases SQLite, MySQL, PostgreSQL and perform CRUD operations like Create, Read, Update, Delete. (BTL6)

Assessment Tool Weightage: 35%

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 20**

Question Count: 2

Code of Conduct

I HARIHARAN G(192424156)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: HARIHARAN G

Database Design and Connectivity Rubric

Requirement Analysis (4 Marks)	ER Diagram Design (4 Marks)	Table Design & Normalization (4 Marks)	Database Connectivity using Python (4 Marks)	CRUD Operations (4 Marks)	Total (20 Marks)

Total Marks Awarded:

Questions:

Set A

Question 3: Online Shopping Database (10 Marks)

An e-commerce company uses the following schema:

- Customer (Customer_ID, Customer_Name, City)

- Product (Product_ID, Product_Name, Price)
- Orders (Order_ID, Customer_ID, Product_ID, Quantity, Order_Date)
- Analyze the schema and write a Python program to retrieve the customer name, product name, quantity

Question 4: Hospital Management Database (10 Marks)

A hospital maintains the following database schema:

- Doctor (Doctor_ID, Doctor_Name, Specialization)
- Patient (Patient_ID, Patient_Name, Age)
- Appointment (Appointment_ID, Patient_ID, Doctor_ID, Appointment_Date)

Analyze the schema and write a Python program to retrieve the doctor name, patient name, and appointment date for all appointments scheduled during the current month.

Analyze the schema and write a Python program to retrieve the student name, course name, and semester for all students enrolled in courses carrying more than 3 credits.

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Signature of the Course Faculty

Question 3: Online Shopping Database

Aim

To develop a Python program using the SQLite database that retrieves the customer name, product name, quantity, and order date by joining multiple tables.

Problem Statement

An e-commerce company maintains the following database schema:

- Customer(Customer_ID, Customer_Name, City)
- Product(Product_ID, Product_Name, Price)
- Orders(Order_ID, Customer_ID, Product_ID, Quantity, Order_Date)

Write a Python program to retrieve:

- Customer Name
- Product Name

- Quantity
- Order Date

using SQL JOIN operations.

Theory

SQLite is a lightweight relational database that is integrated with Python through the **sqlite3** library. The database stores related information in multiple tables. Foreign keys are used to establish relationships between tables.

The **INNER JOIN** operation combines records from different tables based on matching values. In this program:

- Customer table stores customer details.
- Product table stores product information.
- Orders table stores order transactions.

The SQL JOIN retrieves meaningful information by combining all three tables.

Algorithm

Step 1: Import the sqlite3 module.

Step 2: Create or connect to the SQLite database.

Step 3: Create Customer, Product, and Orders tables.

Step 4: Insert sample records into each table.

Step 5: Execute an SQL JOIN query.

Step 6: Display customer name, product name, quantity, and order date.

Step 7: Close the database connection.

Python Program

```
import sqlite3


conn = sqlite3.connect("shopping.db")

cur = conn.cursor()


cur.execute("""

CREATE TABLE Customer(

Customer_ID INTEGER PRIMARY KEY,
```

```
Customer_Name TEXT,
City TEXT)
""")

cur.execute("""
CREATE TABLE Product(
Product_ID INTEGER PRIMARY KEY,
Product_Name TEXT,
Price REAL)
""")

cur.execute("""
CREATE TABLE Orders(
Order_ID INTEGER PRIMARY KEY,
Customer_ID INTEGER,
Product_ID INTEGER,
Quantity INTEGER,
Order_Date TEXT)
""")

cur.executemany("INSERT INTO Customer VALUES(?,?,?)",
[
(1,'Nisha','Chennai'),
(2,'Rahul','Coimbatore')
])

cur.executemany("INSERT INTO Product VALUES(?,?,?)",
[
(101,'Laptop',55000),
(102,'Mouse',700)
])
```

```
cur.executemany("INSERT INTO Orders VALUES(?,?,?,?,?)",
[
(1,1,101,1,'2026-08-01'),
(2,2,102,2,'2026-08-02')
])

query = """
SELECT Customer_Name,
Product_Name,
Quantity,
Order_Date
FROM Customer
JOIN Orders
ON Customer.Customer_ID=Orders.Customer_ID
JOIN Product
ON Orders.Product_ID=Product.Product_ID;
"""

print("Customer Name | Product Name | Quantity | Order Date")

for row in cur.execute(query):
    print(row)

conn.commit()
conn.close()
```

Output

```
SELECT Customer_Name,
Product_Name,
Quantity,
Order_Date
FROM Customer
JOIN Orders
ON Customer.Customer_ID=Orders.Customer_ID
JOIN Product
ON Orders.Product_ID=Product.Product_ID;
"""

print("Customer Name | Product Name | Quantity | Order Date")

for row in cur.execute(query):
    print(row)

conn.commit()
conn.close()
```

```
Customer Name | Product Name | Quantity | Order Date
('Nisha', 'Laptop', 1, '2026-08-01')
('Rahul', 'Mouse', 2, '2026-08-02')
```

Result

Thus, the Python program was successfully developed using SQLite to retrieve the customer name, product name, quantity, and order date using SQL JOIN operations.

Question 4: Hospital Management Database

Aim

To develop a Python program using SQLite to retrieve the doctor name, patient name, and appointment date for appointments scheduled during the current month.

Problem Statement

A hospital maintains the following database schema:

- Doctor(Doctor_ID, Doctor_Name, Specialization)
- Patient(Patient_ID, Patient_Name, Age)
- Appointment(Appointment_ID, Patient_ID, Doctor_ID, Appointment_Date)

Write a Python program to retrieve:

- Doctor Name
- Patient Name
- Appointment Date

for appointments scheduled during the current month.

Theory

Hospital management systems maintain information in multiple related tables. The Appointment table acts as a bridge between the Doctor and Patient tables using foreign keys.

SQLite provides built-in date functions such as **date('now')** and **date('now','start of month')** to retrieve records based on the current month.

The SQL JOIN operation combines the Doctor, Patient, and Appointment tables to display complete appointment information.

Algorithm

Step 1: Import sqlite3.

Step 2: Create Hospital database.

Step 3: Create Doctor, Patient, and Appointment tables.

Step 4: Insert sample records.

Step 5: Execute an SQL JOIN query with the current month condition.

Step 6: Display doctor name, patient name, and appointment date.

Step 7: Close the database connection.

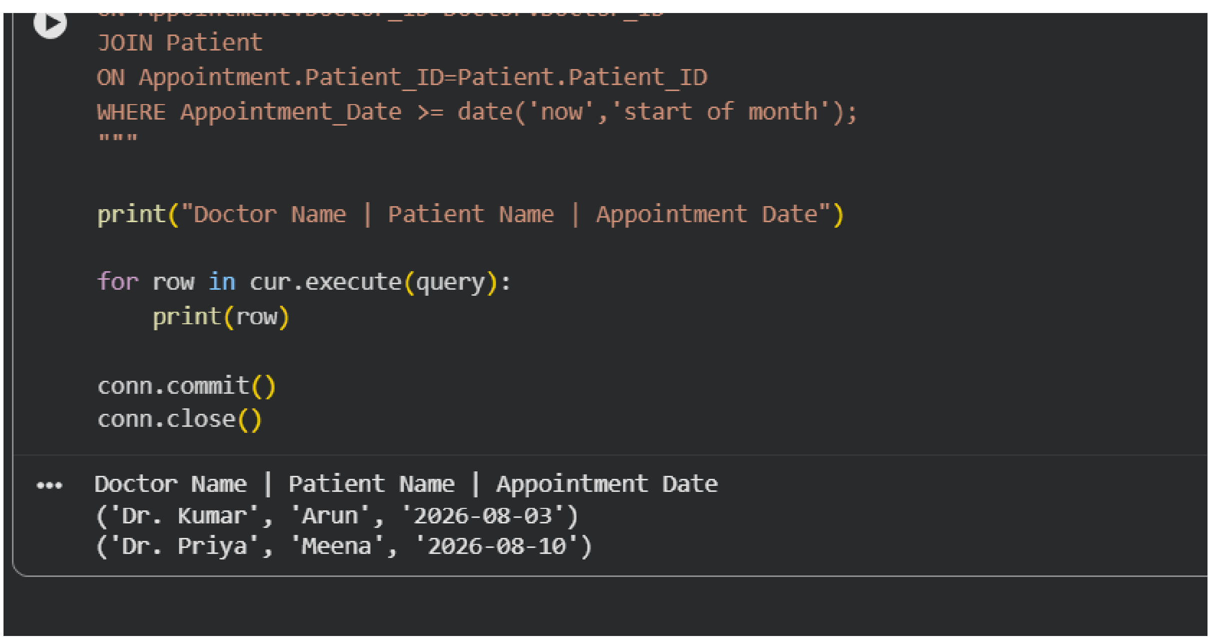
Python Program

```
import sqlite3


conn = sqlite3.connect("hospital.db")

cur = conn.cursor()


cur.execute("""

CREATE TABLE Doctor(

Doctor_ID INTEGER PRIMARY KEY,

Doctor_Name TEXT,

Specialization TEXT)

""")


cur.execute("""
```

```
CREATE TABLE Patient(
Patient_ID INTEGER PRIMARY KEY,
Patient_Name TEXT,
Age INTEGER)
""")

cur.execute("""
CREATE TABLE Appointment(
Appointment_ID INTEGER PRIMARY KEY,
Patient_ID INTEGER,
Doctor_ID INTEGER,
Appointment_Date TEXT)
""")

cur.executemany("INSERT INTO Doctor VALUES(?,?,?)",
[
(1,'Dr. Kumar','Cardiology'),
(2,'Dr. Priya','Neurology')
])

cur.executemany("INSERT INTO Patient VALUES(?,?,?)",
[
(1,'Arun',30),
(2,'Meena',25)
])

cur.executemany("INSERT INTO Appointment VALUES(?,?,?,?)",
[
(1,1,1,'2026-08-03'),
(2,2,2,'2026-08-10')
])
```

```
query = """

SELECT Doctor_Name,

Patient_Name,

Appointment_Date

FROM Appointment

JOIN Doctor

ON Appointment.Doctor_ID=Doctor.Doctor_ID

JOIN Patient

ON Appointment.Patient_ID=Patient.Patient_ID

WHERE Appointment_Date >= date('now','start of month');

"""


print("Doctor Name | Patient Name | Appointment Date")


for row in cur.execute(query):

    print(row)


conn.commit()

conn.close()
```

Output



```
JOIN Patient
ON Appointment.Patient_ID=Patient.Patient_ID
WHERE Appointment_Date >= date('now','start of month');
"""

print("Doctor Name | Patient Name | Appointment Date")

for row in cur.execute(query):
    print(row)

conn.commit()
conn.close()
```

```
Doctor Name | Patient Name | Appointment Date
('Dr. Kumar', 'Arun', '2026-08-03')
('Dr. Priya', 'Meena', '2026-08-10')
```

Result

Thus, the Python program was successfully developed using SQLite to retrieve the doctor name, patient name, and appointment date for appointments scheduled during the current month using SQL JOIN operations.